ADR-2026-37

Migration Validation via pg_depend

Status: Accepted | **Date:** 2026-03-11 | **Owner:** Daniel Shanklin / Director of AI & Technology

Related: ADR-2026-38 (Unified Migration Authority), elt-forge Feature Spec

BLUF (Bottom Line Up Front)

When validating migrations, use PostgreSQL's pg_depend for existing object dependencies and AI context reasoning for pending migrations. No SQL parsing. This is how the CI gate in cerebro-migrations knows whether a migration stack is safe to merge.

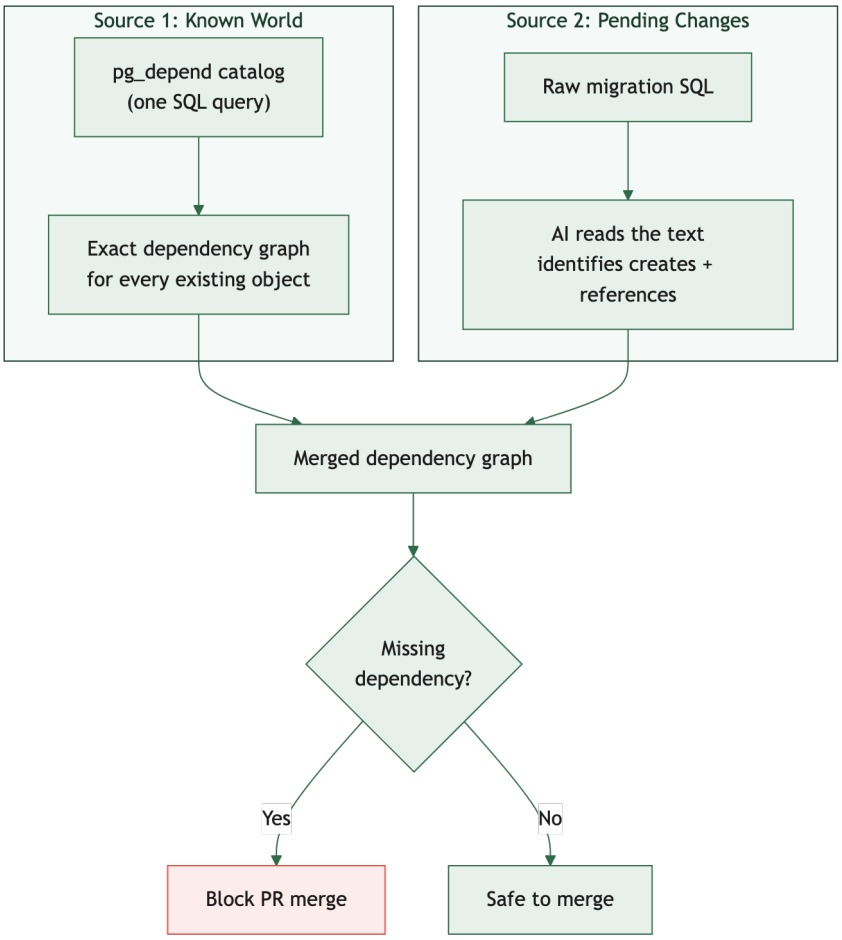
> **DECISION: Use pg_depend (ground truth for existing objects) + AI context reasoning (for pending migrations) to validate migration stacks in CI. No SQL parsing. No manifest files.**

The Problem

Before a migration merges, we need to answer: 'will this migration succeed against the real database?' That means knowing what depends on what.

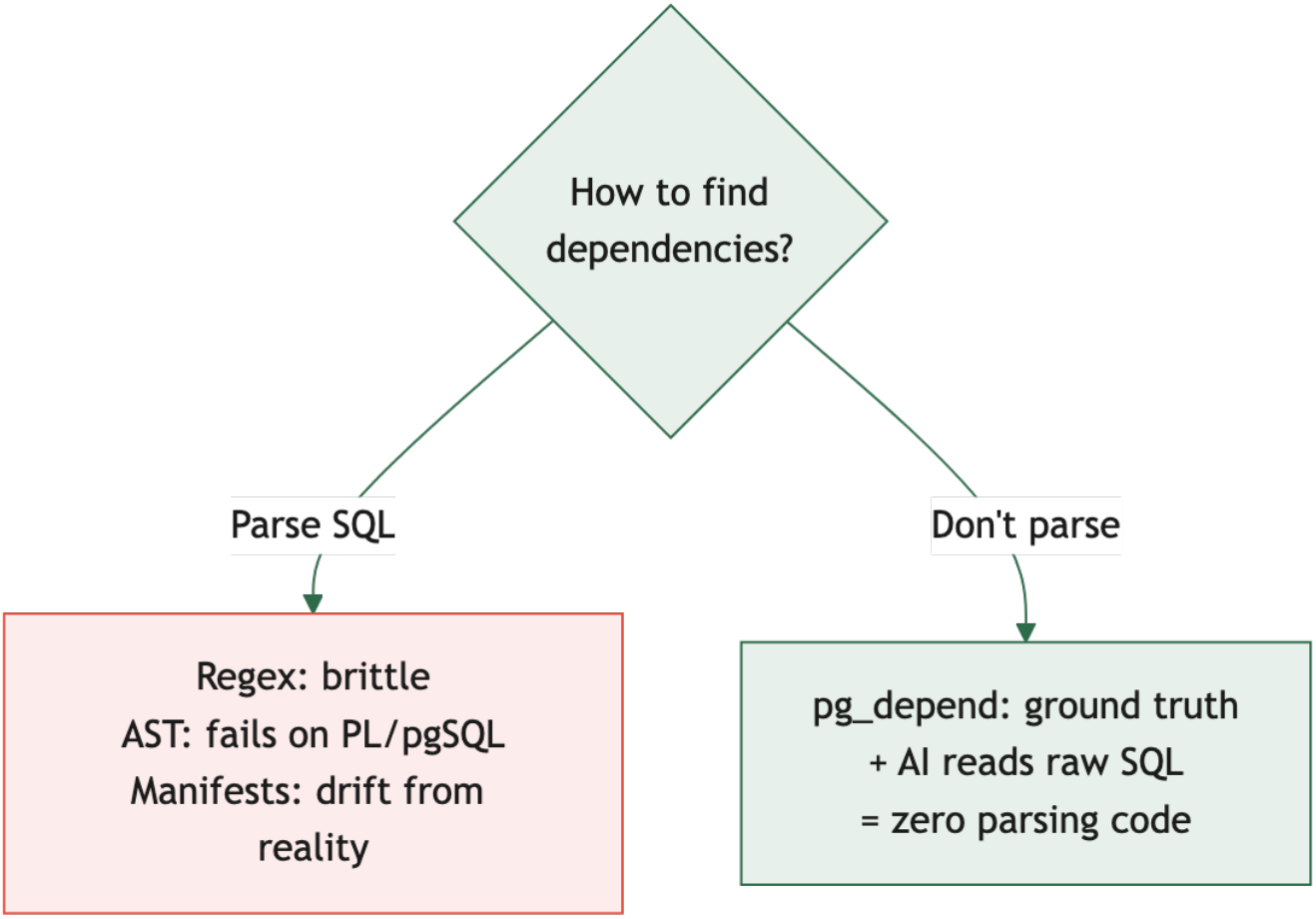
What we need to know	Where the answer lives
What objects exist today and how they relate	PostgreSQL's **pg_depend** catalog — ground truth
What a pending migration will create and reference	The **raw SQL text** of the migration file

The Decision: Two-Source Dependency Resolution



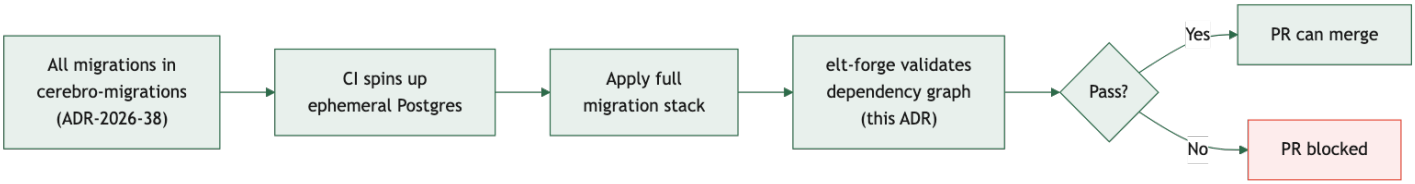
pg_depend handles the known world with zero ambiguity. AI context reasoning handles pending migrations that don't exist in the catalog yet. The merge of both gives a complete picture with zero parsing code to maintain.

Why Not Parse SQL?



Rejected Approach	Why
Regex parsing	Brittle. Breaks on dynamic DDL, PL/pgSQL, conditional creates.
Full AST parser	Fails on Supabase extensions. Heavy dependency for marginal gain.
Manifest/header files	Developer burden. Drifts from reality. Another thing to sync.
pg_depend alone	Can't see pending migrations. Only knows current state.
Context reasoning alone	Works but no ground-truth anchor. May miss subtle dependencies.

How It Fits Together



ADR-2026-38 answers where migrations live and who owns them. **This ADR** answers how we validate them. Together they form the complete migration safety system.

When to Revisit

- If schema count exceeds ~50 and context window pressure becomes real
- If a production failure traces back to AI misreading a pending migration
- If PostgreSQL changes pg_depend semantics (unlikely — stable since 8.0)